Ruby

Ruby is a text-based language that offers a great progression when moving on from Scratch. Primarily designed to be programmer-friendly, it has a syntax that's close to English.

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Background

Ruby was released in 1995 by Japanese computer scientist Yukihiro Matsumoto, who wanted to design a simple and general-purpose scripting language. Matsumoto wanted his language to implement all the features necessary to write in an object-oriented style. Ruby was designed to make it easier for programmers to do tasks, rather than making it easier for computers to run code quickly.

```
[irb(main):007:0> puts "Hello, World!"
Hello, World!
=> nil
irb(main):008:0>
```

```
[irb(main):014:0> apples = 3
=> 3
[irb(main):015:0> oranges = 4
=> 4
[irb(main):016:0> fruit = apples + oranges
=> 7
irb(main):018:0>
```

The prompt appears at the start of each line.

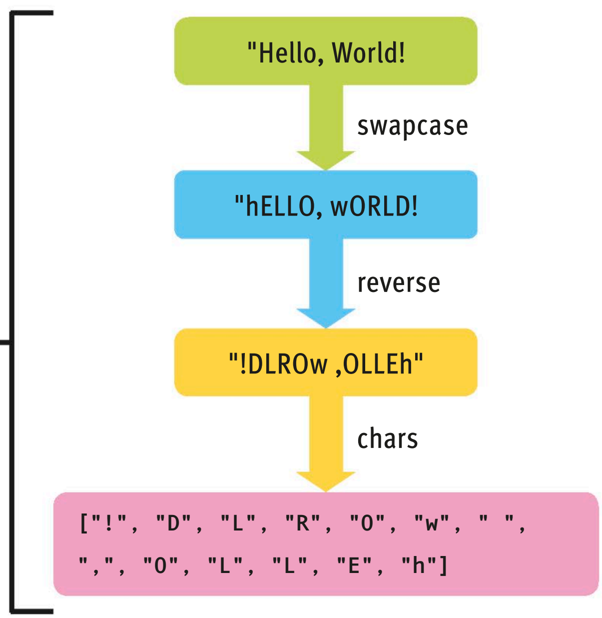
Result of the command just executed

▷ **Interactive Ruby**
Ruby has an interactive interpreter, called the Interactive Ruby Shell or IRB. It allows programmers to type individual commands that can be executed immediately.

How does it work?

Ruby has a very different approach from most languages. Almost everything in it is an object, even items like numbers and characters. These objects have methods that allow programmers to do things with them. In Ruby, it's also possible to program in a mostly imperative style. This allows new programmers to gradually get to grips with the object-oriented style.

```
"Hello, World!".swapcase.reverse.chars
```



▷ **Chaining**
Ruby aims to be concise. One example of this is chaining, which applies several methods to an object. The method names are separated by dots and applied from left to right.